ARROWSTACK DATA ANALYTICS INTERNSHIP

Task 2

Advanced Excel Financial Modeling & Pivot Dashboard

Project report using the S&P 500 Financial Dataset in Microsoft Excel (Power Query, PivotTables, PivotCharts and Dashboard).

Item	Details
Prepared by	[Your Name]
Date	04 November 2026
Dataset	financial_data_sp500_companies.csv
Deliverable	Arrowstack_Task2_Advanced_Excel_Financial_Model.xlsx
Tools	Microsoft Excel, Power Query, PivotTables, PivotCharts, GitHub

1. Project Overview

This project analyses the financial performance of S&P 500 companies in Microsoft Excel. The raw dataset was imported and cleaned with Power Query, checked for data quality issues, summarised into headline financial KPIs, analysed company by company with PivotTables and a PivotChart, and finally presented in a single-page dashboard.

Objectives

- Import and clean the S&P 500 financial dataset using Power Query.
- Produce a Financial Data Quality Report.
- Calculate key financial KPIs such as revenue, net income, gross profit and profitability rate.
- Compare company performance using PivotTables and a Top 10 PivotChart.
- Build a financial performance dashboard and document the work on GitHub.

Dataset at a Glance

Attribute	Value
Total records	2,012
Unique companies	502
Source columns	Date, company, ticker, revenue, gross profit, operating income, net income, expenses and other financial fields

2. Dataset Preparation and Cleaning

The CSV file was imported into Excel through Power Query and transformed in the Power Query Editor before being loaded into the workbook.

1. Imported the financial dataset into Excel using Power Query.
2. Opened the data in the Power Query Editor for processing.
3. Promoted the first row to headers and removed unnecessary columns.
4. Renamed columns where required (for example Firm to Company, Total Revenue to Total_Revenue).
5. Corrected report dates using locale-aware date conversion.
6. Converted the financial columns into appropriate numeric data types.
7. Loaded the cleaned dataset into the financial_data worksheet (2,012 rows loaded).

Result: a clean, structured table named financial_data that feeds every other sheet in the workbook.

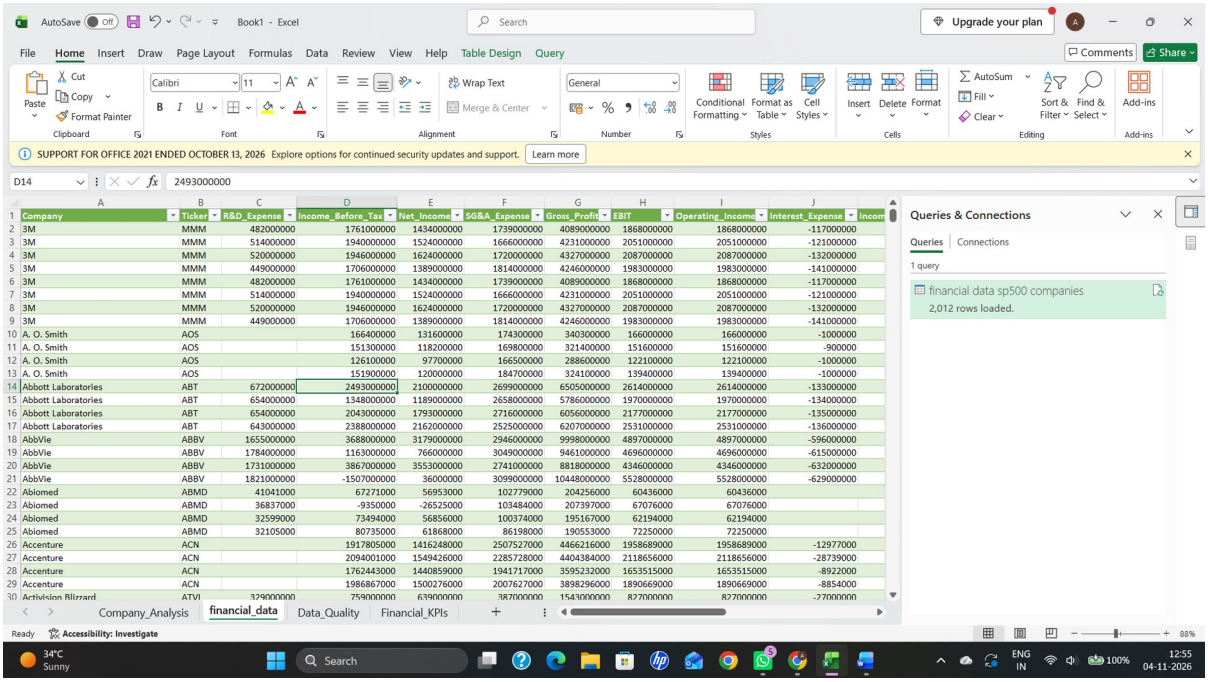


Figure 1: Cleaned dataset loaded in the financial_data sheet (Query: 2,012 rows loaded)

3. Financial Data Quality Report

A separate Data_Quality worksheet was created to validate the cleaned dataset.

Quality Check	Result	Status
Total records	2,012	-
Unique companies	502	-
Missing company names	0	Pass
Missing tickers	0	Pass
Missing report dates	0	Pass
Duplicate company-date records	5	Review
Missing net income	0	Pass
Missing gross profit	0	Pass
Missing revenue	0	Pass

Observation: all key fields are complete. Five company-date records are duplicated and should be reviewed, because duplicates can slightly inflate totals.

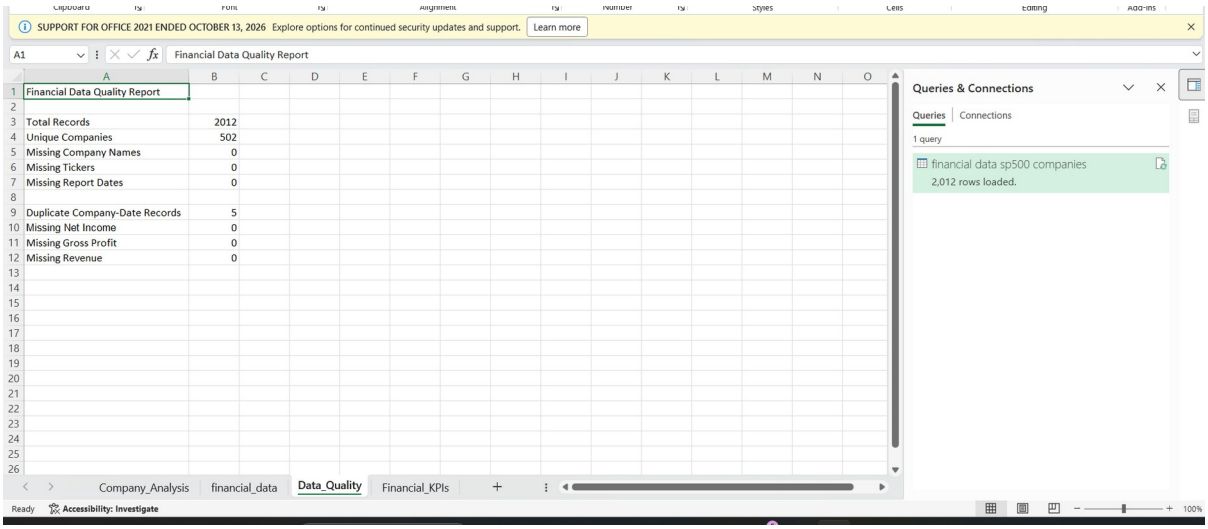
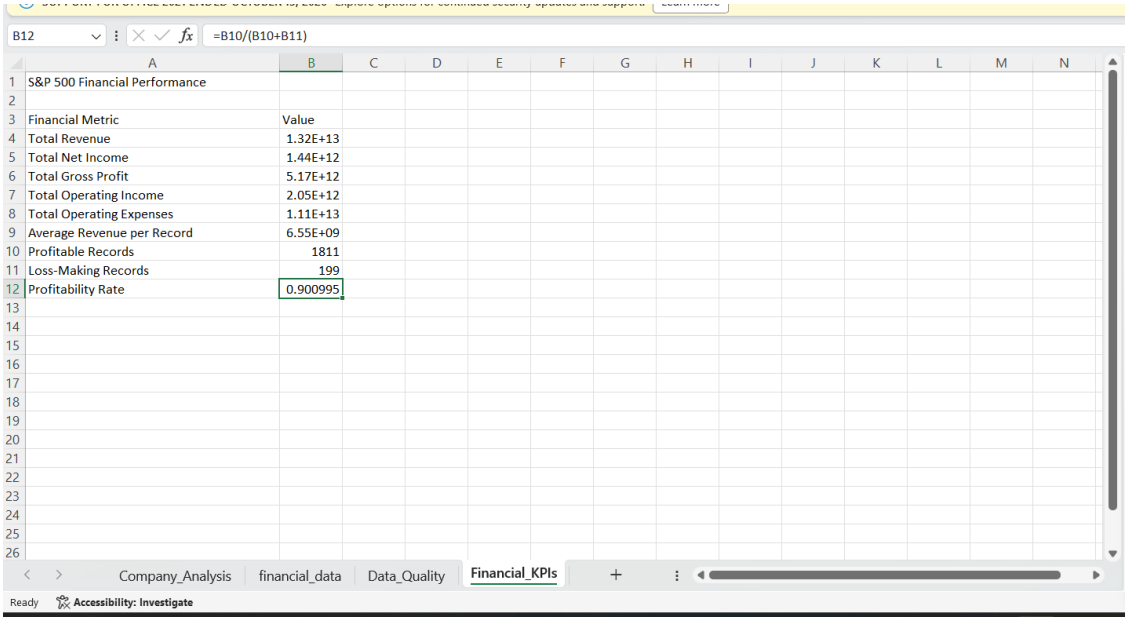


Figure 2: Data_Quality worksheet (Financial Data Quality Report)

4. Financial KPI Analysis

The Financial_KPIs worksheet calculates the headline metrics used by the dashboard. The Profitability Rate is computed as Profitable Records / (Profitable Records + Loss-Making Records), i.e. 1,811 / 2,010.

Financial Metric	Result	Approx. Value
Total Revenue	1.31733E+13	$13.17 trillion
Total Net Income	1.44187E+12	$1.44 trillion
Total Gross Profit	5.16553E+12	$5.17 trillion
Total Operating Income	2.05E+12	$2.05 trillion
Total Operating Expenses	1.11E+13	$11.1 trillion
Average Revenue per Record	6.55E+09	$6.55 billion
Profitable Records	1,811	-
Loss-Making Records	199	-
Profitability Rate	90.10%	-



	A	B
1	S&P 500 Financial Performance	
2		
3	Financial Metric	Value
4	Total Revenue	1.32E+13
5	Total Net Income	1.44E+12
6	Total Gross Profit	5.17E+12
7	Total Operating Income	2.05E+12
8	Total Operating Expenses	1.11E+13
9	Average Revenue per Record	6.55E+09
10	Profitable Records	1811
11	Loss-Making Records	199
12	Profitability Rate	0.900995

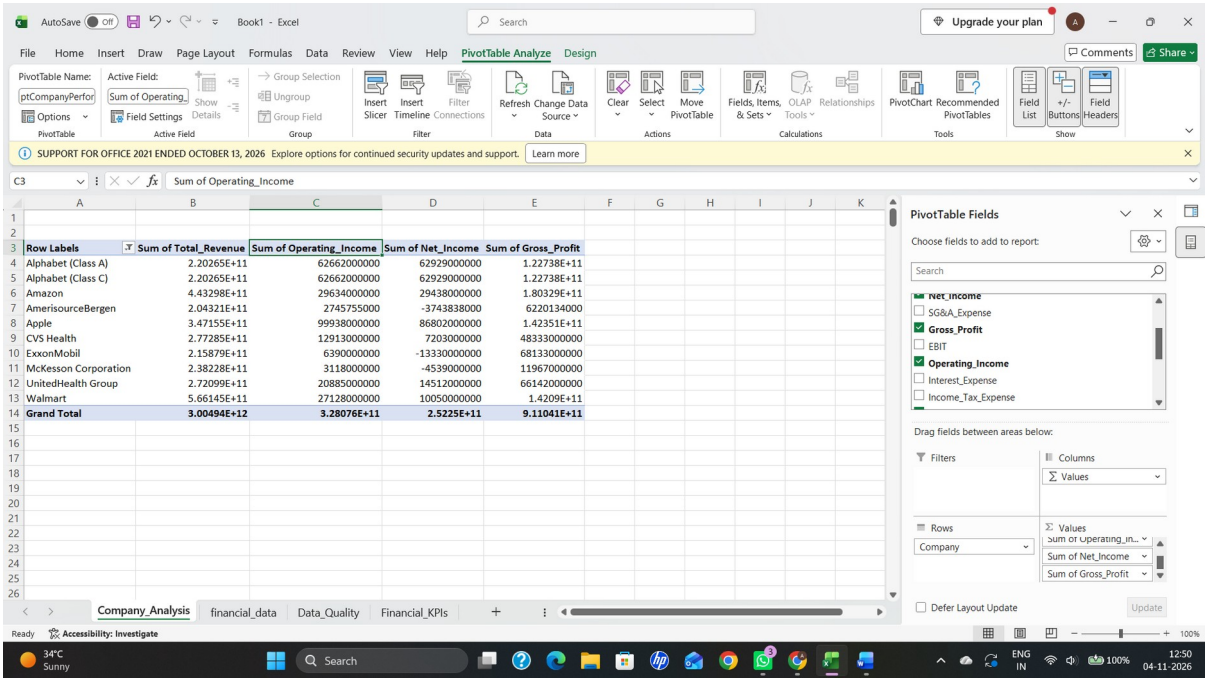
Figure 3: Financial_KPIs worksheet (Profitability Rate formula =B10/(B10+B11))

5. Company-wise Financial Analysis

The Company_Analysis worksheet uses PivotTables with Company in Rows and the following measures in Values:

- Total Revenue by company
- Operating Income by company
- Net Income by company
- Gross Profit by company

A Top 10 Companies Financial Performance PivotChart was added to compare these measures visually.



Row Labels	Sum of Total_Revenue	Sum of Operating_Income	Sum of Net_Income	Sum of Gross_Profit
Alphabet (Class A)	2.20265E+11	62662000000	62929000000	1.22738E+11
Alphabet (Class C)	2.20265E+11	62662000000	62929000000	1.22738E+11
Amazon	4.43298E+11	29634000000	29438000000	1.80329E+11
AmerisourceBergen	2.04321E+11	2745755000	-3743838000	6220134000
Apple	3.47155E+11	99938000000	86802000000	1.42351E+11
CVS Health	2.77285E+11	12913000000	7203000000	48333000000
ExxonMobil	2.15879E+11	6390000000	-13330000000	68133000000
McKesson Corporation	2.38228E+11	3118000000	-4539000000	11967000000
UnitedHealth Group	2.72099E+11	20885000000	14512000000	66142000000
Walmart	5.66145E+11	27128000000	10050000000	1.4209E+11
Grand Total	**3.00494E+12**	**3.28076E+11**	**2.5225E+11**	**9.11041E+11**

Figure 4: Company_Analysis PivotTable with PivotTable Fields pane

Row Labels	Sum of Total_Revenue	Sum of Operating_Income	Sum of Net_Income	Sum of Gross_Profit
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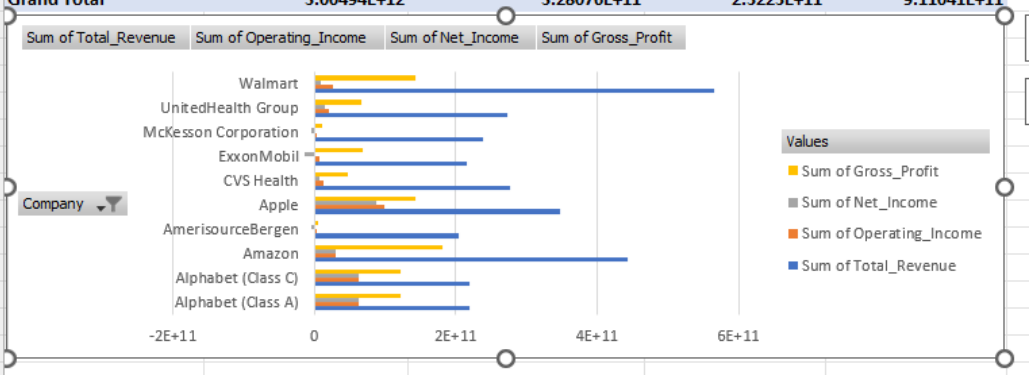


Figure 5: PivotTable and Top 10 Companies PivotChart

Key Observations

- **Walmart** has the highest revenue among the companies shown (about $566 billion).
- **Apple** leads in operating income (about $99.9 billion) and net income (about $86.8 billion).
- **Amazon** is second in revenue (about $443 billion) with a gross profit of about $180 billion.
- **ExxonMobil, AmerisourceBergen and McKesson** show negative net income in the records included.
- Alphabet (Class A) and Alphabet (Class C) report identical figures, as both share classes belong to the same company.

6. Financial Performance Dashboard

A separate Dashboard worksheet presents the results on one page. It contains:

- Dashboard title: S&P 500 Financial Performance Dashboard
- Total Revenue KPI
- Total Net Income KPI
- Total Gross Profit KPI
- Profitability Rate KPI (percentage format)
- Top 10 Companies Financial Performance chart

The KPI cards are linked to the Financial_KPIs sheet by formulas, so they update automatically if the data changes. Formatting added includes the dashboard title, KPI labels, percentage formatting and a chart title.

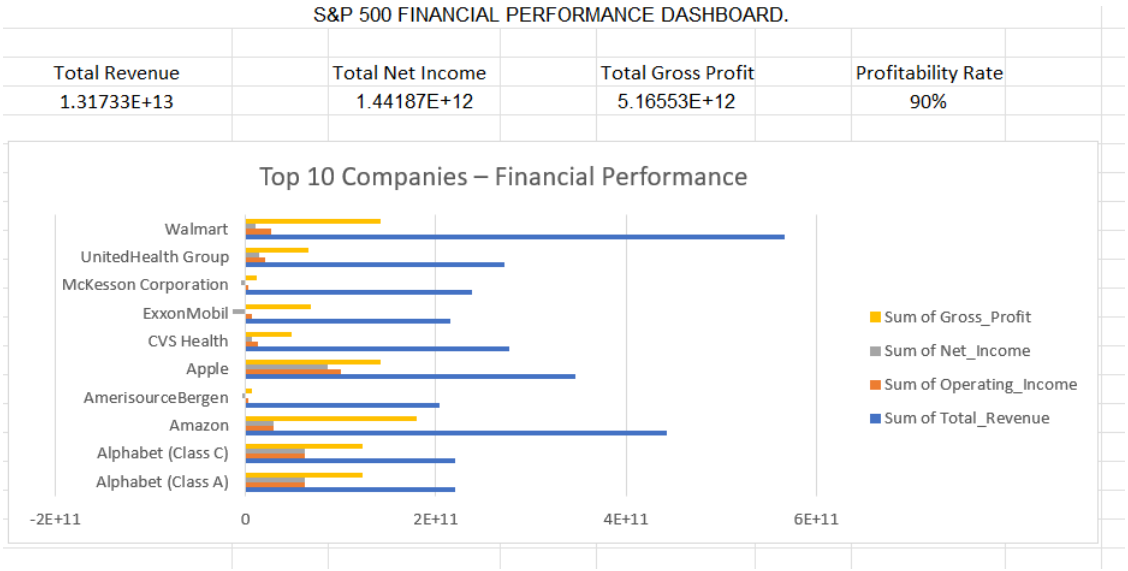


Figure 6: Final Financial Performance Dashboard

7. GitHub Documentation

A GitHub repository named **Advanced Excel Financial Modeling & Pivot Dashboard** was created, and the Excel workbook was uploaded. A detailed README.md was also prepared covering:

- Project overview
- Dataset details
- Data cleaning methodology
- Data quality checks
- Financial KPIs
- PivotTable and PivotChart analysis
- Dashboard explanation
- Workbook structure and instructions

Repository link: [Paste GitHub link]

8. Final Project Structure

Workbook: **Arrowstack_Task2_Advanced_Excel_Financial_Model.xlsx**

Worksheet	Purpose
Dashboard	Financial dashboard with KPI cards and Top 10 chart
Financial_KPIs	KPI calculations
Company_Analysis	PivotTable and PivotChart
Data_Quality	Data validation report
financial_data	Cleaned dataset from Power Query

9. Limitations and Future Improvements

- The Top 10 chart plots revenue, operating income, net income and gross profit together. Their scales differ greatly, so smaller measures are hard to read. Separate charts or a secondary axis would improve clarity.
- KPI values appear in scientific notation (for example 1.31733E+13). Displaying them in billions or trillions would be easier to read.
- Five duplicate company-date records remain flagged and could be removed or reviewed to avoid double counting.
- Adding slicers for company and year would make the dashboard interactive.

10. Conclusion

Task 2 delivered a complete Excel financial modelling workflow: data cleaning in Power Query, a data quality report, financial KPI analysis, company-wise PivotTable and PivotChart analysis, and a Financial Performance Dashboard. The workbook and documentation are published on GitHub as part of the Arrowstack Data Analytics Internship.